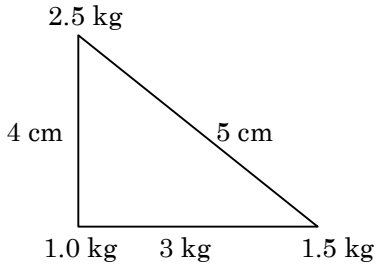
JEE Main Online Exam 2020

Questions & Solutions

7th January 2020 | Shift - I

PHYSICS

Q.1 Three point particles of masses 1.0 kg, 1.5 kg and 2.5 kg are placed at three corners of a right angle triangle of sides 4.0 cm, 3.0 cm and 5.0 cm as shown in the figure. The center of mass of the system is at a point -



(1) 1.5 cm right and 1.2 cm above 1 kg mass

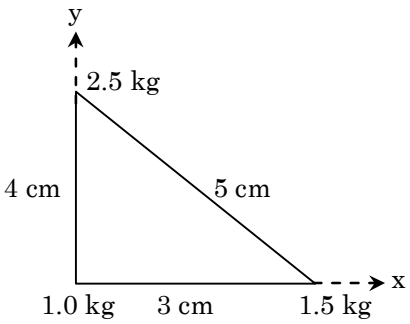
(2) 2.0 cm right and 0.9 cm above 1 kg mass

(3) 0.9 cm right and 2.0 cm above 1 kg mass

(4) 0.6 cm right and 2.0 cm above 1 kg mass

Ans. **[3]**

Sol.



$$x_{cm} = \frac{2.5\times0+1.5\times3+1\times0}{2.5+1+1.5}$$

$$x_{cm} = 0.9 \text{ cm}$$

$$y_{cm} = \frac{2.5\times4+1\times0+1.5\times0}{5}$$

$$y_{cm} = 2.0 \text{ cm}$$

Q.2 Visible light of wavelength 6000×10^{-8} cm falls normally on a single slit and produces a diffraction pattern. It is found that the second diffraction minimum is at 60º from the central maximum. If the first minimum is produced at θ_1, then θ_1 is close to -

(1) 45º (2) 20º (3) 30º (4) 25º

Ans. **[4]**

Sol. For 2nd minima

$d \sin\theta = 2\lambda$

$d \sin 60^\circ = 2\lambda$

$$\Rightarrow \frac{\lambda}{d} = \frac{\sqrt{3}}{4}$$

For 1st minima

$\Rightarrow d \sin\theta = \lambda$

$$\sin\theta = \frac{\lambda}{d} = \frac{\sqrt{3}}{4}$$

$\theta \approx 25^\circ$

Q.3 The time period of revolution of electron in its ground state orbit in a hydrogen atom is 1.6×10^{-16} s. The frequency of revolution of the electron in its first excited state (in s^{-1}) is -

(1) 1.6×10^{14} (2) 5.6×10^{12} (3) 6.2×10^{15} (4) 7.8×10^{14}

Ans. **[4]**

Sol. $$\frac{2\pi r}{v} \propto \frac{n^2/z}{z/n}$$

$$\frac{r}{v} \propto \frac{n^3}{z^2}$$

$$\frac{1}{f} \propto \frac{n^3}{z^2}$$

$$f \propto \frac{z^2}{n^3}$$

$$f_2 = \frac{1}{8} f_1$$

$$= \frac{1}{8} \times \frac{1}{1.6 \times 10^{-16}}$$

$f_2 = 7.8 \times 10^{14}$

Q.4 A litre of dry air at STP expands adiabatically to volume of 3 litres. If $\gamma = 1.40$, the work done by air is : ($3^{1.4} = 4.6555$) [Take air to be an ideal gas]

(1) 100.8 J (2) 48 J (3) 90.5 J (4) 60.7 J

Ans. **[3]**

Sol. $P_iV_i^{\gamma} = P_fV_f^{\gamma}$

$$P_f = \frac{P_i}{3^{\gamma}} = \frac{10^5}{3^{1.4}}$$

$$\Delta w = \frac{P_fV_f - P_iV_i}{1-\gamma}$$

$$= \frac{\frac{10^5}{3^{1.4}} \times 3 \times 10^{-3} - 10^5 \times 10^{-3}}{1-1.4}$$

$\Delta w \approx 90.5$ J

Q.5 A long solenoid of radius R carries a time (t) – dependent current $I(t) = I_0t(1 - t)$. A ring of radius 2R is placed coaxially near its middle. During the time interval $0 \le t \le 1$, the induced current (I_R) and the induced EMF(V_R) in the ring change as -

(1) Direction of I_R remains unchanged and V_R is maximum at t = 0.5

(2) At t = 0.25 direction of I_R reverses and V_R is maximum

(3) Direction of I_R remains unchanged and V_R is zero at t = 0.25

(4) At t = 0.5 direction of I_R reverses and V_R is zero

Ans. **[4]**

Sol. $I = I_0\, t(1 - t)$

$= I_0t - I_0t^2$

$\phi = \mu_0 n\, IA$

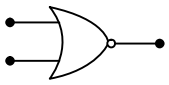
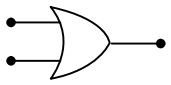
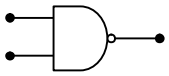
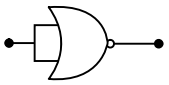
$$V_R = -\frac{d\varphi}{dt} = \mu_0 nA\,(I_0 - 2I_0t)$$

$V_R = 0$ at t = 0.5

$$I_R = \frac{V_R}{\text{Resistance}} = \frac{\mu_0 nA(I_0 - 2I_0t)}{\text{Resistance}}$$

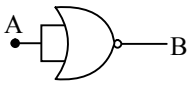
after t = 0.5 I_R, reverses its direction

Q.6 Which of the following gives a reversible operation ?

(1)  (2)  (3)  (4) 

Ans. **[4]**

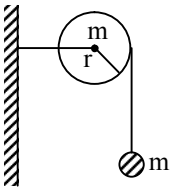
Sol.



A	B
0	1
1	0

It will behave as a NOT gate. Hence gives reversible operation.

Q.7

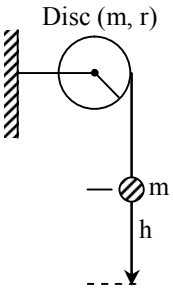


As shown in the figure, a bob of mass m is tied by massless string whose other end portion is wound on a fly wheel (disc) of radius r and mass m. When released from rest the bob starts falling vertically. When it has covered a distance of h, the angular speed of the wheel will be -

(1) $r\sqrt{\frac{3}{2gh}}$ (2) $\frac{1}{r}\sqrt{\frac{4gh}{3}}$ (3) $r\sqrt{\frac{3}{4gh}}$ (4) $\frac{1}{r}\sqrt{\frac{2gh}{3}}$

Ans. **[2]**

Sol.



Energy conservation

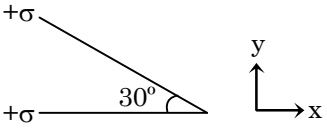
$mgh = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$ ....(1)

$\omega = \frac{v}{r}$ ....(2)

Solving (1) & (2), we get

$\boxed{\omega = \frac{1}{r}\sqrt{\frac{4gh}{3}}}$

Q.8 Two infinite planes each with uniform surface charge density + σ are kept in such a way that the angle between them is 30º. The electric field in the region shown between them is given by -

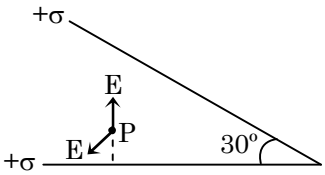


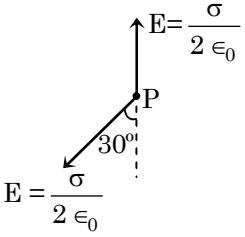
(1) $\frac{\sigma}{2\epsilon_0}\left[(1+\sqrt{3})\hat{y}+\frac{\hat{x}}{2}\right]$ (2) $\frac{\sigma}{\epsilon_0}\left[\left(1+\frac{\sqrt{3}}{2}\right)\hat{y}+\frac{\hat{x}}{2}\right]$

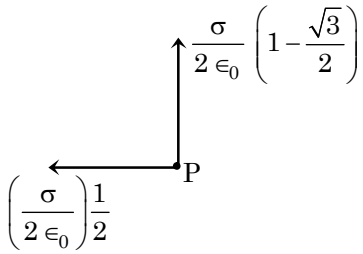
(3) $\frac{\sigma}{2\epsilon_0}\left[(1+\sqrt{3})\hat{y}-\frac{\hat{x}}{2}\right]$ (4) $\frac{\sigma}{2\epsilon_0}\left[\left(1-\frac{\sqrt{3}}{2}\right)\hat{y}-\frac{\hat{x}}{2}\right]$

Ans. **[4]**

Sol.







$$= \frac{\sigma}{2\epsilon_0}\left[\left(1-\frac{\sqrt{3}}{2}\right)\hat{y}-\frac{\hat{x}}{2}\right]$$

Q.9 A polarizer-analyser set is adjusted such that the intensity of light coming out of the analyser is just 10 % of the original intensity. Assuming that the polarizer-analyser set does not absorb any light, the angle by which the analyser need to be rotated further to reduce the output intensity to be zero, is -

(1) 90º (2) 45º (3) 18.4º (4) 71.6º

Ans. **[3]**

Sol. $I = I_0 \cos^2\theta$

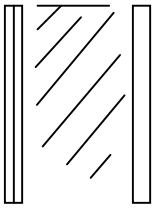
$$\frac{I_0}{10} = I_0 \cos^2\theta$$

$$\cos\theta = \frac{1}{\sqrt{10}}$$

$\theta \approx 71.6$

Hence, further 18.4º rotation reduce output intensity to be zero.

Q.10

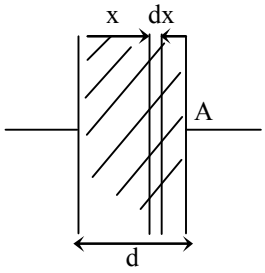


A parallel plate capacitor has plates of area A separted by distance 'd' between them. It is filled with a dielectric which has a dielectric constant that varies as k(x) = K(1 + αx) where 'x' is the distance measured from one of the plates. If (αd) <<1, the total capacitance of the system is best given by the expression :

(1) $\frac{AK\in_0}{d}(1+\alpha d)$ (2) $\frac{A\in_0 K}{d}\left(1+\frac{\alpha^2d^2}{2}\right)$ (3) $\frac{AK\in_0}{d}\left(1+\frac{\alpha d}{2}\right)$ (4) $\frac{A\in_0 K}{d}\left(1+\left(\frac{\alpha d}{2}\right)^2\right)$

Ans. **[3]**

Sol. k(x) = K(1 + αx)



$dC_1 = \frac{K\in_0 A}{dx}$

$dC_1 = \frac{K(1+\alpha x)A\in_0}{dx}$

$\frac{1}{C_{eq}} = \int\frac{1}{dC_1}$

$= \int_0^d \frac{dx}{KA(1+\alpha x)\in_0}$

$\frac{1}{C_{eq}} = \frac{1}{\in_0 KA\alpha}\ \ell n\,(1+\alpha d)$

Using expansion of $\ell n\,(1+x)$ keeping x << 1

$\frac{1}{C_{eq}} = \frac{1}{\in_0 KA\alpha}\left(\alpha d-\frac{\alpha^2d^2}{2}\right)$

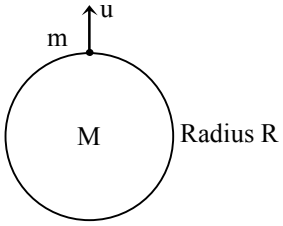
$C_{eq} = \frac{\in_0 KA}{d}\left(1-\frac{\alpha d}{2}\right)^{-1} = \frac{\in_0 KA}{d}\left(1+\frac{\alpha d}{2}\right)$

Q.11 A satellite of mass m is launched vertically upwards with an initial speed u from the surface of the earth. After it reaches height R(R = radius of the earth), it ejects a rocket of mass $\frac{m}{10}$ so that subsequently the satellite moves in a circular orbit. The kinetic energy of the rocket is (G is the gravitational constant; M is the mass of the earth) :

(1) $5m\left(u^2-\frac{119}{200}\frac{GM}{R}\right)$ (2) $\frac{m}{20}\left(u-\sqrt{\frac{2GM}{3R}}\right)^2$ (3) $\frac{3m}{8}\left(u+\sqrt{\frac{5GM}{6R}}\right)^2$ (4) $\frac{m}{20}\left(u^2+\frac{113}{200}\frac{GM}{R}\right)$

Ans. **[1]**

Sol.



Using energy conservation

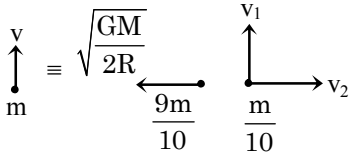
$$\frac{1}{2}mu^2-\frac{GMm}{R}=-\frac{GMm}{2R}+\frac{1}{2}mv^2$$

$$u^2-\frac{2GM}{R}=-\frac{GM}{R}+v^2$$

$$u^2-\frac{GM}{R}=v^2$$

Again after ejecting rocket

$$\vec{P}_i=\vec{P}_f$$



$$\frac{9m}{10}\sqrt{\frac{GM}{2R}}=\frac{m}{10}v_2\ (\vec{p}\ \text{conservation along tangent})$$

$$V_2=9\sqrt{\frac{GM}{2R}}$$

$$mv=\frac{m}{10}v_1\ (\vec{p}\ \text{conservation along radius})$$

$$v_1=10\,v$$

$$\text{K.E. of rocket}=\frac{1}{2}\frac{m}{10}(v_1^2+v_2^2)$$

$$=\frac{1}{2}\frac{m}{10}\left(100\left(u^2-\frac{GM}{k}\right)+81\frac{GM}{2R}\right)$$

$$=\frac{m}{20}\left(100u^2-\frac{100GM}{R}+\frac{81GM}{2R}\right)$$

$$\boxed{\text{K.E.}=5m\left(u^2-\frac{119}{200}\frac{GM}{R}\right)}$$

Q.12 A LCR circuit behaves like a damped harmonic oscillator. Comparing it with a physical spring-mass damped oscillator having damping constant 'b', the correct equivalence would be -

(1) $L \leftrightarrow k$, $C \leftrightarrow b$, $R \leftrightarrow m$

(2) $L \leftrightarrow m$, $C \leftrightarrow k$, $R \leftrightarrow b$

(3) $L \leftrightarrow m$, $C \leftrightarrow \frac{1}{k}$, $R \leftrightarrow b$

(4) $L \leftrightarrow \frac{1}{b}$, $C \leftrightarrow \frac{1}{m}$, $R \leftrightarrow \frac{1}{k}$

Ans. **[3]**

Sol. Spring mass damped oscillator

$$\frac{d^2x}{dt^2} + \frac{b}{m}\frac{dx}{dt} + \frac{k}{m}x = 0 \qquad \ldots.(1)$$

For LC oscillations

$$L\frac{di}{dt} + iR + \frac{q}{C} = 0$$

$$\frac{d^2q}{dt^2} + \frac{R}{L}\frac{dq}{dt} + \frac{q}{LC} = 0 \qquad \ldots.(2)$$

Comparing (1) & (2)

$b \leftrightarrow R$

$L \leftrightarrow m$

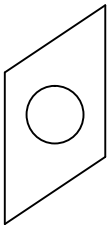
$\frac{1}{C} \leftrightarrow K$

Q.13 Consider a circular coil of wire carrying constant current I, forming a magnetic dipole. The magnetic flux through an infinite plane that contains the circular coil and excluding the circular coil area is given by ϕ_i. The magnetic flux through the area of the circular coil area is given by ϕ_0. Which of the following option is correct ?

(1) $\phi_i < \phi_0$ (2) $\phi_i > \phi_0$ (3) $\phi_i = \phi_0$ (4) $\phi_i = -\phi_0$

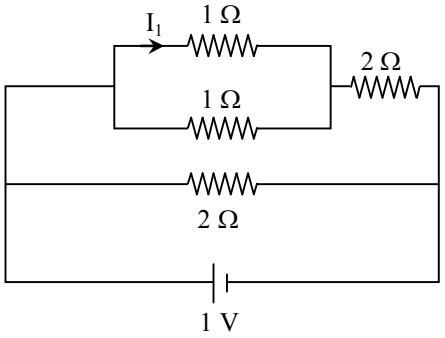
Ans. **[4]**

Sol.



As, magnetic field lines forms a closed loop, hence each line from circular area will pass through outer area in opposite direction hence $\phi_i = -\phi_0$.

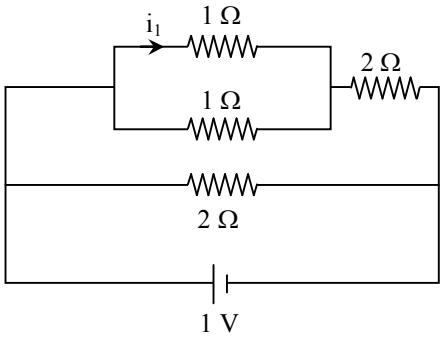
Q.14 The current I_1 (in A) flowing through 1Ω resistor in the following circuit is –



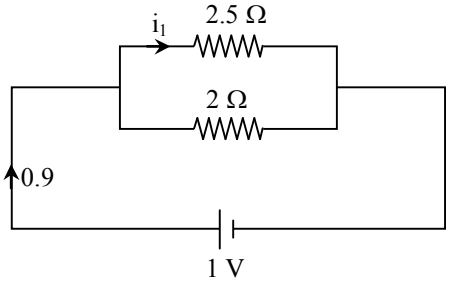
(1) 0.2 (2) 0.4 (3) 0.25 (4) 0.5

Ans. **[1]**

Sol.



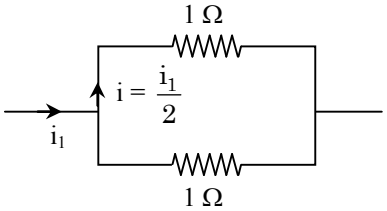
$\Downarrow$



$$I = \frac{V}{R_{eq}} = 1\left(\frac{2}{5}+\frac{1}{2}\right)$$

$$\boxed{I = 0.9\ A}$$

$$i_1 = \frac{I}{4.5}\times 2$$



$$\boxed{i = \frac{I}{4.5} = 0.2\ A}$$

Q.15 Speed of a transverse wave on a straight wire (mass 6.0 g, length 60 cm and area of cross-section 1.0 mm^2) is 90 ms^{-1}. If the Young's modulus of wire is 16×10^{11} Nm^{-2}, the extension of wire over its natural length is -

(1) 0.01 mm (2) 0.04 mm (3) 0.02 mm (4) 0.03 mm

Ans. **[4]**

Sol. $v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{TA}{A\mu}}$

$\text{Stress} = \frac{T}{A} = Y \times \text{Strain} = \frac{Y\Delta L}{L}$

$\Rightarrow \quad v = \sqrt{\frac{Y\Delta L}{L}\frac{A}{m/L}} \quad \left(\because \quad \mu = \frac{m}{L}\right)$

$v = \sqrt{\frac{Y\Delta LA}{m}}$

$\Delta L = \frac{v^2 m}{YA}$

$= \frac{90 \times 90 \times 6 \times 10^{-3}}{16 \times 10^{11} \times 1 \times 10^{-6}}$

$\boxed{\Delta L \approx 0.03 \text{ mm}}$

Q.16 If the magnetic field in a plane electromagnetic wave is given by $\vec{B} = 3 \times 10^{-8} \sin(1.6 \times 10^3 x + 48 \times 10^{10} t)\ \hat{j}$ T, then what will be expression for electric field ?

(1) $\vec{E} = [9\sin(1.6 \times 10^3 x + 48 \times 10^{10} t)\ \hat{k}$ V/m]

(2) $\vec{E} = [60 \sin(1.6 \times 10^3 x + 48 \times 10^{10} t)\ \hat{k}$ V/m]

(3) $\vec{E} = [3 \times 10^{-8} \sin(1.6 \times 10^3 x + 48 \times 10^{10} t)\ \hat{i}$ V/m]

(4) $\vec{E} = [3 \times 10^{-8} \sin(1.6 \times 10^3 x + 48 \times 10^{10} t)\ \hat{j}$ V/m]

Ans. **[1]**

Sol. $E_0 = B_0 c$

$= (3 \times 10^{-8})(3 \times 10^8) \sin(\omega t + kx)$

$E_0 = 9 \sin(1.6 \times 10^3 x + 48 \times 10^{10} t)$

Q.17 If we need a magnification of 375 from a compound microscope of tube length 150 mm and an objective of focal length 5 mm, the focal length of the eye-piece, should be close to -

(1) 12 mm (2) 33 mm (3) 22 mm (4) 2 mm

Ans. **[3]**

Sol. M.P. = 375

$f_0 = 5$mm

Case-I

If final image is at inifinity

$$M.P. = \frac{1}{f_0}\left(\frac{D}{f_e}\right)$$

$$375 = \frac{150}{5}\left(\frac{250}{f_e}\right)$$

$\boxed{f_e = 20}$

Case-2

Near – point adjustment

$$M.P. = \frac{L_0}{f_0}\left(1+\frac{D}{f_e}\right)$$

$$375 = \frac{150}{5}\left(1+\frac{250}{f_e}\right)$$

$f_e \approx 22$ mm

Q.18 A 60 HP electric motor lifts an elevator having a maximum total load capacity of 2000 kg. If the frictional force on the elevator is 4000 N, the speed of the elevator at full load is close to : (1 HP = 746 W, g = 10 ms^{-2})

(1) 1.7 ms^{-1} (2) 1.9 ms^{-1} (3) 2.0 ms^{-1} (4) 1.5 ms^{-1}

Ans. **[2]**

Sol. P = Fv + Mgv

Applied power = 4000 × v + 20000 v

60 × 246 = 4000 v + 20000 v

v = 1.865

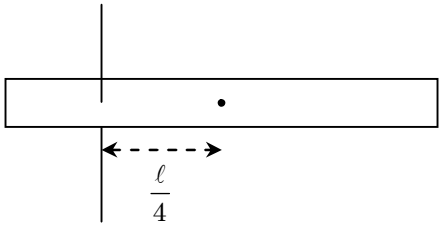
$\boxed{v \approx 1.9 \text{ m/s}}$

Q.19 The radius of gyration of a uniform rod of length ℓ, about an axis passing through a point $\frac{\ell}{4}$ away from the centre of the rod, and perpendicular to it, is -

(1) $\sqrt{\frac{3}{8}}\ell$ (2) $\frac{1}{8}\ell$ (3) $\sqrt{\frac{7}{48}}\ell$ (4) $\frac{1}{4}\ell$

Ans. **[3]**

Sol. Radius of gyration



$$I = \frac{M\ell^2}{12} + M\left(\frac{\ell}{4}\right)^2$$

$$I = \frac{7}{48}M\ell^2$$

$$Mk^2 = \frac{7}{48}M\ell^2$$

$\boxed{k = \ell\sqrt{\frac{7}{48}}}$

Q.20 Two moles of an ideal gas with $\frac{C_P}{C_V}=\frac{5}{3}$ are mixed with 3 moles of another ideal gas with $\frac{C_P}{C_V}=\frac{4}{3}$. The value of $\frac{C_P}{C_V}$ for the mixture is -

(1) 1.50 (2) 1.42 (3) 1.47 (4) 1.45

Ans. **[2]**

Sol. $\gamma_{mix}=\frac{n_1C_{p_1}+n_2C_{p_2}}{n_1C_{v_1}+n_2C_{v_2}}$

$C_p=\left(\frac{\gamma}{\gamma-1}\right)R$

$C_v=\frac{R}{\gamma-1}$

$\gamma_{mix}=\frac{n_1\frac{\gamma_1}{\gamma_1-1}+\frac{n_2\gamma_2}{\gamma_2-1}}{\frac{n_1}{\gamma_1-1}+\frac{n_2}{\gamma_2-1}}$

$n_1=2, \gamma_1=\frac{5}{3}, n_2=3, \gamma_2=\frac{4}{3}$

We get

$\gamma_{mix}=1.42$

Q.21 A beam of electromagnetic radiation of intensity 6.4×10^{-5} W/cm^2 is comprised of wavelength, λ = 310 nm. It falls normally on a metal (work function ϕ = 2eV) of surface area of 1 cm^2. If one in 10^3 photons ejects an electron, total number of electrons ejected in 1 s is 10^x. (hc = 1240 eVnm, 1eV = 1.6×10^{-19} J), then x is ……..... .

Ans. **[11]**

Sol. Energy of photon $=\frac{1240}{310}=4$eV

Energy is greater than work function

No. of photons $=\frac{\text{Intensity}}{\text{Energy of one photon}}$

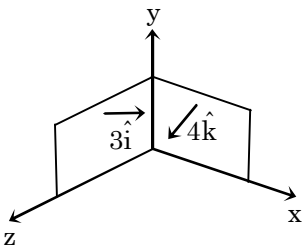
$=\frac{6.4\times10^{-5}}{4\times1.6\times10^{-19}}=10^{14}$

Total no. of electron $=\frac{10^{14}}{10^3}=10^{11}$

Q.22 A loop ABCDEFA of straight edges has six corner points A(0, 0, 0), B(5, 0, 0), C(5, 5, 0), D(0, 5, 0), E(0, 5, 5) and F(0, 0, 5). The magnetic field in this region is $\vec{B} = (3\hat{i} + 4\hat{k})$ T. The quantity of flux through the loop ABCDEFA (in Wb) is …… .

Ans. **[175]**

Sol.



Area of plane = 25

$\phi = (3 \times 25) + 4 \times 25 = 175$ weber

Q.23 A non-isotropic solid metal cube has coefficients of linear expansion as : 5×10^{-5}/ºC along the x-axis and 5×10^{-6}/ºC along the y and the z-axis. If the coefficient of volume expansion of the solid is $C \times 10^{-6}$/ºC then the value of C is …… .

Ans. **[60]**

Sol. <u>For non-isotropic</u>

$\gamma = \alpha_x + \alpha_y + \alpha_z$

$= 5 \times 10^{-5} + 5 \times 10^{-6} + 5 \times 10^{-6}$

$= 60 \times 10^{-6}$

Comparing, we get

$\boxed{C = 60}$

Q.24 A Carnot engine operates between two reservoirs of temperatures 900 K and 300 K. The engine performs 1200 J of work per cycle. The heat energy (in J) delivered by the engine to the low temperature reservoir, in a cycle, is …….. .

Ans. **[600]**

Sol. $\eta = 1 - \frac{T_L}{T_H}$

$= 1 - \frac{300}{900}$

$\frac{w}{Q_s} = \frac{2}{3}$

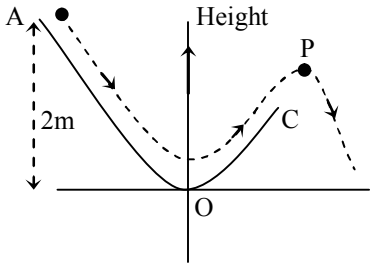
$Q_s = \frac{3}{2}\ w$

$= \frac{3}{2} \times 1200$

$Q_s = 1800$ J

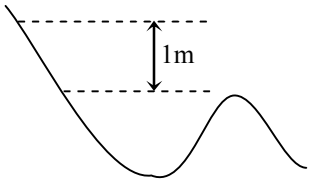
Rejected = $Q_s - w = 1800 - 1200 = 600$ J

Q.25 A particle (m = 1 kg) slides down a frictionless track (AOC) starting from rest at a point A (height 2 m). After reaching C, the particle continues to move freely in air as a projectile. When it reaching its highest point P (height 1 m), the kinetic energy of the particle (in J) is : (Figure drawn is schematic and not to scale; take g = 10 ms^{-2}) …….. .



Ans. **[10]**

Sol.



K.E. = ΔP.E.

= mgh

= 1 × 10 × 1 = 10 Joule.

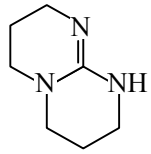
JEE Main Online Exam 2020

Question & Solutions

7th January 2020 | Shift - I

CHEMISTRY

Q.1 The increasing order of pK_b for the following compounds will be

(A) $NH_2–CH=NH$, (B)  (C) CH_3NHCH_3

(1) (B) < (C) < (A) (2) (C) < (A) < (B) (3) (B) < (A) < (C) (4) (A) < (B) < (C)

Ans. **[2]**

Sol. $pK_b \propto \frac{1}{K_b} \propto \frac{1}{Basic\ strength}$

(B) > (A) > (C)

Q.2 The IUPAC name of the complex $[Pt(NH_3)_2Cl(NH_2CH_3)]Cl$ is :

(1) Bisammine(methanamine)chloridoplatinum(II)chloride
(2) Diammine(methanamine)chloridoplatinum(II)chloride
(3) Diamminechlorido(methanamine)platinum(II)chloride
(4) Diamminechlorido(aminomethane)platinum(II)chloride

Ans. **[3]**

Sol. $[Pt(NH_3)_2Cl(NH_2CH_3)]Cl$

$x + 0 – 1 + 0 = +1$

$x = +2 \rightarrow Pt^{+2}$

Diamminechlorido(methanamine)platinum(II)chloride

Q.3 Given that the standard potentials (E^o) of Cu^{2+}/Cu and Cu+/Cu are 0.34 V and 0.522 V respectively, the E^o of Cu^{2+}/Cu^+ is -

(1) + 0.158 V (2) 0.182 V (3) – 0.158 V (4) – 0.182 V

Ans. **[1]**

Sol. $e– + Cu^{+2} \xrightarrow{E_1^0} Cu^{\oplus} \xrightarrow{0.522V} Cu$

0.34

$2 \times 0.34 = E_1^0 + 1 \times 0.522$

$E_1^0 = 0.68 – 0.522$

$E_1^0 = 0.158$

Q.4 Consider the following reactions :

(a) $(CH_3)_3CCH(OH)CH_3 \xrightarrow{cons.\ H_2SO_4}$

(b) $(CH_3)_2CHCH\ (Br)CH_3 \xrightarrow{cons.\ KOH}$

(c) $(CH_3)_2CHCH(Br)CH_3 \xrightarrow{(CH_3)_3O^{\ominus}K^{\oplus}}$

(d) $(CH_3)_2\underset{\underset{OH}{|}}{C}-CH_2-CHO \xrightarrow{\Delta}$

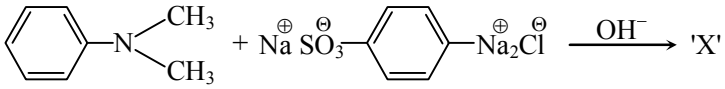
Which of these reactions(s) will not produce Saytzeff product ?

(1) (d) only (2) (c) only (3) (b) and (d) (4) (a), (c) and (d)

Ans. **[2]**

Sol. Bulky base always prefer Hoffmann product

Q.5 Consider the following reaction :



The product 'X' is used

(1) as food grade colourant

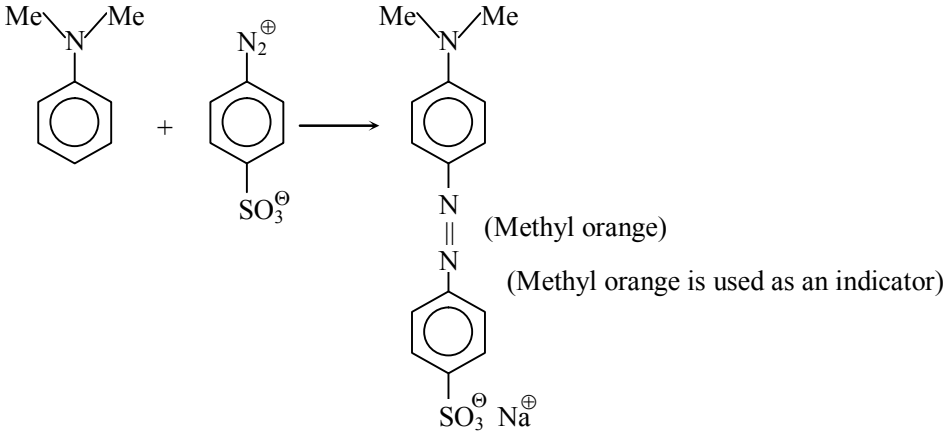
(2) in laboratory test for phenols

(3) in protein estimation as an alternative to ninhydrin

(4) in acid base titration as an indicator

Ans. **[4]**

Sol.



Q.6 Amongst the following statements, that which was not proposed by Dalton was -

(1) All the atoms of a given element have identical properties including identical mass. Atoms of different elements differ in mass

(2) When gases combine or reproduced in a chemical reaction they do so in a simple ratio by volume provided all gases are at the same T & P

(3) Matter consists of indivisible atoms

(4) Chemical reactions involve reorganization of atoms. These are neither created nor destroyed in a chemical reaction.

Ans. **[2]**

Sol. Theory based

Q.7 At 35ºC, the vapour pressure of CS_2 is 512 mm Hg and that of acetone is 344 mm Hg. A solution of CS_2 in acetone has a total vapour pressure of 600 mm Hg. The false statement amongst the following is :

(1) A mixture of 100 mL CS_2 and 100 mL acetone has a volume < 200 mL
(2) Roult's law is not obeyed by this system
(3) CS_2 and acetone are less attracted to each other than to themselves
(4) Heat must be absorbed in order to produce the solution at 35ºC

Ans. **[1]**

Sol. The above mixture will show positive deviation from Raoults law

Q.8 The purest form of commercial iron is :

(1) Cast iron (2) Scarp iron and pig iron (3) Pig iron (4) Wrought iron

Ans. **[4]**

Sol. Theory based

Q.9 The electron gain enthalpy (in kJ/mol) of fluorine, chlorine, bromine and iodine, respectively, are :

(1) – 333, – 349, – 325 and – 296 (2) – 296, – 325, – 333 and – 349
(3) – 333, – 325, – 349 and – 296 (4) – 349, – 333, – 325 and – 296

Ans. **[1]**

Sol. Electron gain enthalpy increases with electro negativity chlorine has higher electron gain enthalpy than fluorine (exception)

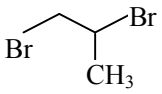
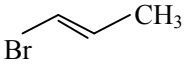
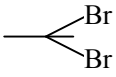
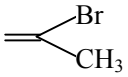
Q.10 In comparison to the zeolite process for the removal of permanent hardness, the synthetic resins method is :

(1) More efficient as it can exchange only cations
(2) Less efficient as the resins cannot be regenerated
(3) More efficient as it can exchange both cations as well as anions
(4) Less efficient as it exchanges only anions.

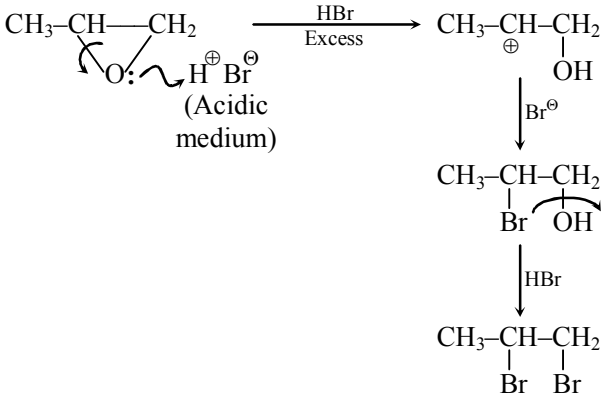
Ans. **[1]**

Sol. Theory based

Q.11 1-Methyl ethylene oxide when treated with an excess of HBr produces :

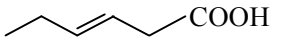
(1) 
(2) 
(3) 
(4) 

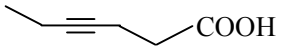
Ans. **[1]**

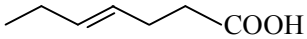
Sol. 

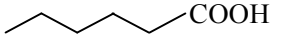
Q.12 What is the product of following reactions ?

$$Hex-3-ynal \xrightarrow[(ii)\ PBr_3\\(iii)\ Mg/ether\\(iv)\ CO_2/H_3O^+]{(i)\ NaBH_4}$$

(1) 

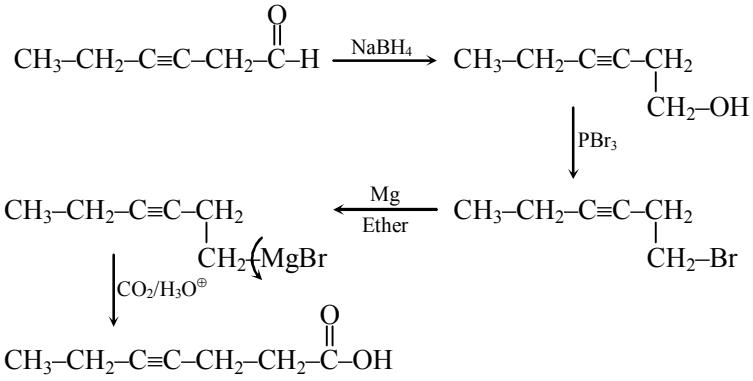
(2) 

(3) 

(4) 

Ans. **[3]**

Sol.



Q.13 The relative strength of interionic / intermolecular forces in decreasing order is

(1) ion-dipole > ion-ion > dipole-dipole

(2) dipole-dipole > ion-dipole > ion-ion

(3) ion-ion > ion-dipole > dipole-dipole

(4) ion-dipole > dipole-dipole > ion-ion

Ans. **[3]**

Sol. Theory based

Q.14 The theory that can completely/properly explain the nature of bonding in $[Ni(Co)_4]$ is

(1) Molecular orbital theory

(2) Valence bond theory

(3) Werner's theory

(4) Crystal field theory

Ans. **[1]**

Sol. $Ni(O)_4$ banding diagram and energy can be explained by MOT clearly

Q.15 Oxidation number of potassium in K_2O, K_2O_2 and KO_2, respectively, is :

(1) +1, +2 and +4 (2) +2, +1 and $+\frac{1}{2}$

(3) +1, +1 and +1 (4) +1, +4 and +2

Ans. **[3]**

Sol. $\underset{2x-2=0}{K_2O} \longrightarrow x=+1$

$\underset{2x-2=0}{K_2O_2} \longrightarrow x=+1$

$\underset{x-1=0}{KO_2} \longrightarrow x=+1$

Q.16 The atomic radius of Ag is closest to :

(1) Ni (2) Hg (3) Au (4) Cu

Ans. **[3]**

Sol. Theory based

Q.17 Match the following :

(i) Riboflavin	(a) Beriberi
(ii) Thiamine	(b) Scurvy
(iii) Pyridoxine	(c) Calosis
(iv) Ascorbic acid	(d) Spasm

(1) (i) – (d), (ii) – (b), (iii) – (a), (iv) – (c)

(2) (i) – (a), (ii) – (d), (iii) – (c), (iv) – (b)

(3) (i) – (c), (ii) – (a), (iii) – (d), (iv) – (b)

(4) (i) – (c), (ii) – (d), (iii) – (a), (iv) – (b)

Ans. **[3]**

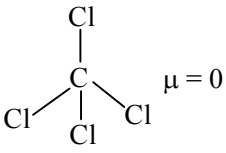
Sol. Theory based

Q.18 The dipole moments of CCl_4, $CHCl_3$ and CH_4 are in the order :

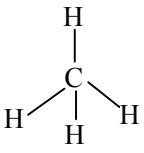
(1) $CCl_4 < CH_4 < CHCl_3$ (2) $CHCl_3 < CH_4 = CCl_4$ (3) $CH_4 = CCl_4 < CHCl_3$ (4) $CH_4 < CCl_4 < CHCl_3$

Ans. **[3]**

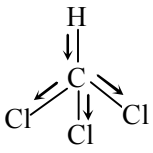
Sol.



$\mu = 0$



$\mu = 0$



$\mu \neq 0$

Q.19 A solution of m-chloroaniline, m-chlorophenol and m-chlorobenzoic acid in ethyl acetate was extracted initially with a saturated solution of $NaHCO_3$ to give fraction A. The left over organic phase was extracted with dilute NaOH solution to give fraction B. The final organic layer was labelled as fraction C. Fraction A, B and C, contain respectively :

(1) m-chlorobenzoic acid, m-chloroaniline and m-chlorophenol

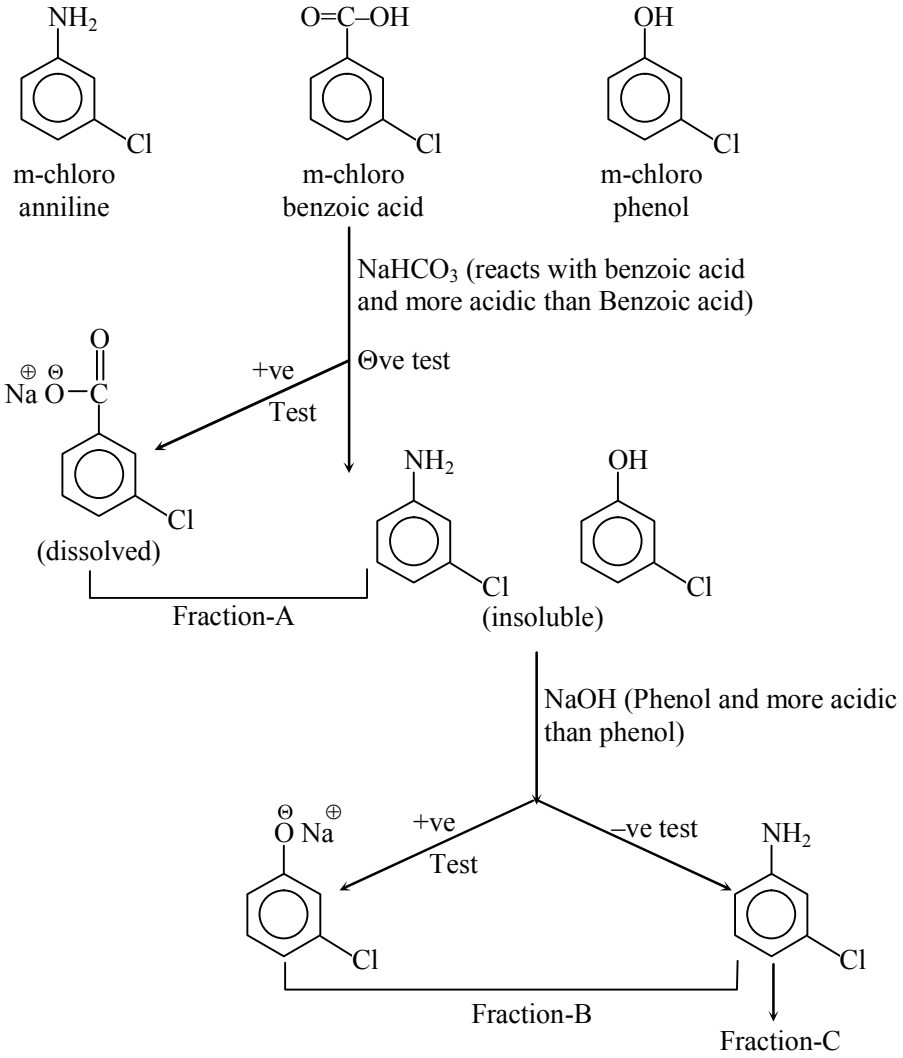
(2) m-chlorophenol, m-chlorobenzoic acid and m-chloroaniline

(3) m-chloroaniline, m-chlorobenzoic acid and m-chlorophenol

(4) m-chlorobenzoic acid, m-chlorophenol and m-chloroaniline

Ans. **[4]**

Sol.



Q.20 The number of orbitals associated with quantum numbers n = 5, $m_s = +\frac{1}{2}$ is :

(1) 15 (2) 50 (3) 11 (4) 25

Ans. **[4]**

Sol.

n=5 $\ell = n - 1$

$= 4$

0 1 2 3 4

$m \rightarrow -\ell$ to $+\ell$

$\rightarrow 0 \rightarrow 0$

$\rightarrow 1 \rightarrow -1, 0, 1$

$\rightarrow 2 \rightarrow -2, -1, 0, 1, 2$

$\rightarrow 3 \rightarrow -3, -2, -1, 0, 1, 2, 3$

$\rightarrow 4 \rightarrow -4, -3, -2, -1, 0, 1, 2, 3, 4$

Total 25 orbitals are possible

Q.21 For the reaction;

$A(l) \longrightarrow 2B(g)$

$\Delta U = 2.1$ kcal, $\Delta S = 20$ cal K^{-1} at 300 K

Hence ΔG in kcal is _____.

Ans. **[– 2.70]**

Sol. $\Delta H = \Delta U + \Delta ngRT$

$= 2.1 \times 10^3 + 2(2)(300) = 3300$ calories

$\Delta G = \Delta H - T\Delta S$

$= 3300 - 300(20) = -2700$ calories

$= -2.7$ Kcal

Q.22 Two solutions, A and B, each of 100 L was made by dissolving 4 g of NaOH and 9.8 g of H_2SO_4 in water, respectively. The pH of the resultant solution obtained from mixing 40 L of solution A and 10 L of solution B is _____.

Ans. **[10.60]**

Sol. $M_{(H_2SO_4)} = \dfrac{9.8}{98 \times 100} = 10^{-3}$ m

$M_{NaOH} = \dfrac{4}{40 \times 100} = 10^{-3}$ m

Equivalents of resulting solution $[OH^-] = \dfrac{40 \times 10^{-3} - 10 \times 10^{-3} \times 2}{50} = \dfrac{2}{5} \times 10^{-3}$

pOH = 3.39 | pH = 10.6

Q.23 During the nuclear explosion, one of the products is ^{90}Sr with half life of 6.93 years. If 1 µg of ^{90}Sr was absorbed in the bones of a newly born baby in place of Ca, how much tine, in years, is required to reduce it by 90% if it is not lost metabolically _____.

Ans. **[23.03]**

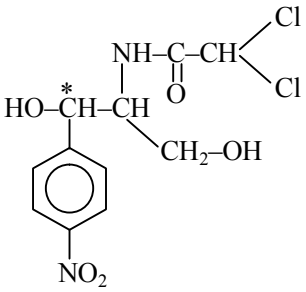
Sol. $\dfrac{t_{90\%}}{t_{50\%}} = \dfrac{\ell n \dfrac{100}{10}}{\ell n 2}$

$t_{90\%} = \dfrac{\ell n_{10}}{\ell n_2} \times t_{50\%} = \dfrac{6.93 \times \ell n 10}{0.693} = 23.03$ years

Q.24 The number of chiral carbons in chloramphenicol is _____.

Ans. **[2.00]**

Sol.



Q.25 Chlorine reacts with hot and concentrated NaOH and produces compounds (X) and (Y). Compound (X) gives white precipitate with silver nitrate solution. The average bond order between Cl and O atoms in (Y) is _____.

Ans. **[1.67]**

Sol. $3Cl_2 + 6\ NaOH \longrightarrow 5\ NaCl + NaClO_3 + 3H_2O$

$NaCl + AgNO_3 \longrightarrow \underset{\text{White ppt}}{AgCl} + NaNO_3$

Y is $NaClO_3 \longrightarrow ClO_3^-$

Bond order = $\frac{5}{3} = 1.67$

CAREER POINT

JEE Main Online Exam 2020

Questions & Solutions

7th January 2020 | Shift - I

MATHEMATICS

Q.1 If $y(\alpha) = \sqrt{2\left(\frac{\tan\alpha+\cot\alpha}{1+\tan^2\alpha}\right)+\frac{1}{\sin^2\alpha}}$, $\alpha \in \left(\frac{3\pi}{4}, \pi\right)$, then $\frac{dy}{d\alpha}$ at $\alpha = \frac{5\pi}{6}$ is :

(1) $\frac{4}{3}$ (2) $-\frac{1}{4}$ (3) -4 (4) 4

Ans. **[4]**

Sol. $y(\alpha) = \sqrt{2\left(\frac{\tan\alpha+\cot\alpha}{1+\tan^2\alpha}\right)+\frac{1}{\sin^2\alpha}}$

$= \sqrt{\frac{2\frac{1}{\sin\alpha\cos\alpha}}{\sec^2\alpha}+\operatorname{cosec}^2\alpha}$

$= \sqrt{2\cot\alpha+\operatorname{cosec}^2\alpha}$

$= |1+\cot\alpha| \quad \alpha \in \left(\frac{3\pi}{4}, \pi\right)$

$y = -1-\cot\alpha$

$\left.\frac{dy}{d\alpha}\right|_{\alpha=\frac{5\pi}{6}} = \operatorname{cosec}^2\alpha$

Q.2 If $\operatorname{Re}\left(\frac{z-1}{2z+i}\right) = 1$, where $z = x + iy$, then the point (x, y) lies on a :

(1) circle whose diameter is $\frac{\sqrt{5}}{2}$. (2) circle whose centre is at $\left(-\frac{1}{2}, -\frac{3}{2}\right)$

(3) straight line whose slope is $-\frac{2}{3}$ (4) straight line whose slope is $\frac{3}{2}$

Ans. **[1]**

Sol. $\operatorname{Re}\left(\frac{z-1}{2z+i}\right) = 1$

put $z = x + iy$

$\operatorname{Re}\left(\frac{x+iy-1}{2(x+iy)+i}\right) = 1$

$\Rightarrow \frac{2x(x-1)+y(2y+1)}{(2x)^2+(2y+1)^2} = 1$

$\Rightarrow$ on solving we get

$x^2 + y^2 + x + \frac{3}{2}y + \frac{1}{2} = 0$

centre $= \left(-\frac{1}{2}, -\frac{3}{4}\right)$

radius $= \sqrt{\frac{1}{4}+\frac{9}{16}-\frac{1}{2}} = \frac{\sqrt{5}}{4}$

diameter $= \frac{\sqrt{5}}{2}$

Q.3 If the distance between the foci of an ellipse is 6 and the distance between its directrices is 12, then the length of its latus rectum is

(1) $2\sqrt{3}$ (2) $\frac{3}{\sqrt{2}}$ (3) $\sqrt{3}$ (4) $3\sqrt{2}$

Ans. **[4]**

Sol. distance b/w focii = 2ae = 6

$\Rightarrow$ ae = 3 …….(i)

distance b/w directrices $= \frac{2a}{e} = 12$

$\frac{a}{e} = 6$ …..(ii)

from (1) and (2)

$a^2 = 18$

also $a^2e^2 = 9$ (from (1))

$18\, e^2 = 9$

$e^2 = \frac{1}{2} = 1 - \frac{b^2}{a^2}$

$b^2 = 9$

$\therefore$ length of L.R. $= \frac{2b^2}{a}$

$= \frac{2\times 9}{\sqrt{18}}$

$\Rightarrow 3\sqrt{2}$

Q.4 Total number of 6-digit numbers in which only and all the five digits 1, 3, 5, 7 and 9 appear, is

(1) $\frac{5}{2}(6!)$ (2) $\frac{1}{2}(6!)$ (3) 5^6 (4) 6!

Ans. **[1]**

Sol. Given digits 1, 3, 5, 7, 9 total six digit numbers with given constraints are

$= {}^5C_1 \times \frac{6!}{2!} = 1800$

Q.5 The greatest positive integer k, for which $49^k + 1$ is a factor of the sum $49^{125} + 49^{124} + \ldots + 49^2 + 49 + 1$, is :

(1) 65 (2) 60 (3) 32 (4) 63

Ans. **[4]**

Sol. $1 + 49 + 49^2 + \ldots\ldots 49^{125}$

sum of G.P. $= \frac{1.\left((49)^{126} - 1\right)}{49 - 1}$

D. $\frac{\left((49)^{63} + 1\right)\left((49)^{63} - 1\right)}{48}$

$\because (49)^{63} - 1$ is divisible by 48

$\therefore k = 63$

Q.6 Let α and β be two real roots of the equation $(k + 1)\tan^2 x - \sqrt{2}\,.\,\lambda \tan x = (1 - k)$, where $k (\neq -1)$ and λ are real numbers. If $\tan^2(\alpha + \beta) = 50$, then a value of λ is :

(1) 5 (2) 10 (3) $5\sqrt{2}$ (4) $10\sqrt{2}$

Ans. **[2]**

Sol. $(k + 1)\tan^2 x - \sqrt{2}\,\lambda \tan x = (1 - k) \begin{matrix}\alpha \\ \beta\end{matrix}$

$\therefore (k + 1)t^2 - \sqrt{2}\,\lambda t = (1 - k) \begin{matrix}\tan\alpha \\ \tan\beta\end{matrix}$

$\tan\alpha + \tan\beta = \frac{\sqrt{2}\lambda}{k+1}$

$\tan\alpha \,.\, \tan\beta = \frac{k-1}{k+1}$

$\tan^2(\alpha + \beta) = \left[\frac{\tan\alpha + \tan\beta}{1 - \tan\alpha.\,\tan\beta}\right]^2 = 50$

$= \left[\frac{\frac{\sqrt{2}\lambda}{k+1}}{1 - \frac{k-1}{k+1}}\right]^2 = 50$

$= \frac{\lambda^2}{2} = 50$

$\Rightarrow \lambda = 10$

Q.7 The logical statement $(p \Rightarrow q) \wedge (q \Rightarrow \sim p)$ is equivalent to :

(1) p (2) ~q (3) q (4) ~p

Ans. **[4]**

Sol. $(p \Rightarrow q) \wedge (q \Rightarrow \sim p)$

p	q	$p \to q$	$\sim p$	$q \to \sim p$	$(p \Rightarrow q) \wedge (p \Rightarrow \sim q)$
T	T	T	F	F	F
T	F	F	F	T	F
F	T	T	T	T	T
F	F	T	T	T	T

$\therefore (p \to q) \wedge (p \to \sim q) = \sim p$

Q.8 If $y = mx + 4$ is a tangent to both the parabolas, $y^2 = 4x$ and $x^2 = 2by$, then b is equal to

(1) –64 (2) –128 (3) –32 (4) 128

Ans. **[2]**

Sol. Given $y = mx + 4$ is tangent to both the parabolas

$\therefore$ Applying condition of tangent

for $y^2 = 4x$

$\Rightarrow 4 = \frac{1}{m} \Rightarrow m = \frac{1}{4}$

for $x^2 = 2by$ line $y = \frac{x}{4} + 4$ is tangent

$\therefore$ solve $x^2 = 2b\left(\frac{x}{4} + 4\right)$

$D = 0$

we get $b^2 + 32 \times 4b = 0$

$b = -128$

Q.9 If $y = y(x)$ is the solution of the differential equation, $e^y\left(\frac{dy}{dx} - 1\right) = e^x$ such that $y(0) = 0$, then $y(1)$ is equal to :

(1) 2e (2) $2 + \log_e 2$ (3) $1 + \log_e 2$ (4) $\log_e 2$

Ans. **[3]**

Sol. $e^y\left(\frac{dy}{dx} - 1\right) = e^x$

$\frac{dy}{dx} - 1 = e^{x-y}$

$x - y = t$

$1 - \frac{dy}{dx} = \frac{dt}{dx}$

$\therefore\ -\frac{dt}{dx} = e^t$

$-\int e^{-t}dt = \int dx$

$e^{-t} = x + c$

$e^{y-x} = x + c$

$f(0) = 0$

$\therefore\ c = 1$

$\therefore\ y = x + \ell n\,(x + 1)$

$f(1) = 1 + \ell n2$

Q.10 If $f(a + b + 1 - x) = f(x)$, for all x, where a and b are fixed positive real numbers, then $\frac{1}{a+b}\int_a^b x(f(x)+f(x+1))dx$ is equal to :

(1) $\int_{a+1}^{b+1} f(x)dx$ (2) $\int_{a-1}^{b-1} f(x)dx$ (3) $\int_{a+1}^{b+1} f(x+1)dx$ (4) $\int_{a-1}^{b-1} f(x+1)dx$

Ans. **[4]**

Sol. $I = \frac{1}{(a+b)}\int_a^b x\big(f(x)+f(x+1)\big)dx$ ...(1)

$x \to a + b - x$

$I = \frac{1}{(a+b)}\int_a^b (a+b-x)\ (f(a+b-x) + f(a+b-x+1))\,dx$

$= \frac{1}{(a+b)}\int_a^b (a+b-x)\ (f(x) + f(x+1))\,dx$ ...(2)

(1) + (2)

$2I = \int_a^b \big(f(x)+f(x+1)\big)\,dx$

$2I = \int_a^b f(x+1)\,dx + \int_a^b f(x)\,dx$

$= \int_a^b f(a+b+1-x)\,dx + \int_a^b f(x)\,dx$

$I = \int_a^b f(x)\,dx$

$x \to x + 1$

$\therefore\ I = \int_{a-1}^{b-1} f(x+1)\,dx$

Q.11 Let the function, $f : [-7, 0] \to R$ be continuous on $[-7, 0]$ and differentiable on $(-7, 0)$. If $f(-7) = -3$ and $f'(x) \le 2$, for all $x \in (-7, 0)$, then for all such functions f, $f(-1) + f(0)$ lies in the interval :

(1) $(-\infty, 20]$ (2) $[-6, 20]$ (3) $[-3, 11]$ (4) $(-\infty, 11]$

Ans. **[1]**

Sol. Using LMVT on $[-7, 0]$

$$\frac{f(0)-f(-7)}{0-(-7)} \le 2$$

$$\Rightarrow f(0) \le 11$$

Using LMVT on $[-7, -1]$

$$\frac{f(-1)-f(-7)}{-1-(-7)} \le 2$$

$$\Rightarrow f(-1) \le 9$$

$$\therefore f(0) + f(-1) \le 20$$

Q.12 Let P be a plane passing through the points (2, 1, 0) , (4, 1, 1) and (5, 0, 1) and R be any point (2, 1, 6). Then the image of R in the plane P is

(1) (4, 3, 2) (2) (6, 5, 2) (3) (6, 5, –2) (4) (3, 4, –2)

Ans. **[3]**

Sol. Equation of plane $= \begin{vmatrix} x-2 & y-1 & z-0 \\ 4-2 & 1-1 & 1-0 \\ 5-2 & 0-1 & 1-0 \end{vmatrix} = 0$

$= x + y - 2z = 3$

$\therefore$ image of R(2, 1, 6) in this plane is

$$\frac{x-2}{1} = \frac{y-1}{1} = \frac{z-6}{-2} = \frac{-2(2+1-12)}{6}$$

$\therefore (x, y, z) = (6, 5, -2)$

Q.13 If $g(x) = x^2 + x - 1$ and $(gof)(x) = 4x^2 - 10x + 5$, then $f\left(\frac{5}{4}\right)$ is equal to:

(1) $-\frac{3}{2}$ (2) $-\frac{1}{2}$ (3) $\frac{3}{2}$ (4) $\frac{1}{2}$

Ans. **[2]**

Sol. $g(f(x)) = f^2(x) + f(x) - 1$ ....(1)

$$gof\left(\frac{5}{4}\right) = 4\left(\frac{5}{4}\right)^2 - 10 \times \frac{5}{4} + 5$$

$$= -\frac{5}{4} \quad \ldots(2)$$

from (1) & (2)

$$gof\left(\frac{5}{4}\right) = f^2\left(\frac{5}{4}\right) + f\left(\frac{5}{4}\right) - 1 = -\left(\frac{5}{4}\right)$$

$$\therefore f\left(\frac{5}{4}\right) = -\frac{1}{2}$$

Q.14 Let α be a root of the equation $x^2 + x + 1 = 0$ and the matrix $A = \frac{1}{\sqrt{3}}\begin{bmatrix} 1 & 1 & 1 \\ 1 & \alpha & \alpha^2 \\ 1 & \alpha^2 & \alpha^4 \end{bmatrix}$, then the matrix A^{31} is equal to

(1) I_3 (2) A^3 (3) A^2 (4) A

Ans. **[2]**

Sol. $x^2 + x + 1 = 0 \begin{cases} \omega \\ \omega^2 \end{cases}$

$$A = \frac{1}{\sqrt{3}}\begin{bmatrix} 1 & 1 & 1 \\ 1 & \alpha & \alpha^2 \\ 1 & \alpha^2 & \alpha^4 \end{bmatrix}$$

put $\alpha = \omega$

$$\therefore A^2 = A.A = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega^4 \end{bmatrix}\begin{bmatrix} 1 & 1 & 1 \\ 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega^4 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$\Rightarrow A^4 = A^2 A^2 = I$

$\therefore A^{31} = A^{28} A^3$

$= A^3$

Q.15 Five number are in A.P. , whose sum is 25 and product is 2520. If one of these five numbers is $-\frac{1}{2}$, then the greatest number amongst them is

(1) 7 (2) 16 (3) $\frac{21}{2}$ (4) 27

Ans. **[2]**

Sol. Let numbers are

$a - 2d , a - d, a, (a + d), a + 2d$

sum $= 5a = 25 \Rightarrow a = 5$

product $(a^2 - 4d^2)(a^2 - d^2) . a = 2520$

$(25 - 4d^2)(25 - d^2) 5 = 2520$

$\Rightarrow 4d^4 - 121 d^2 - 4d^2 + 121 = 0$

$(d^2 - 1)(4d^2 - 121) = 0$

$d^2 = 1 , d^2 = \frac{121}{4}$

$d = \pm 1 , d = \pm \frac{11}{2}$

from $d = \pm 1$ we do not get $-\frac{1}{2}$ as a term

$\therefore d = \frac{11}{2}$ for largest term

$\therefore$ largest term $= 5 + 2d$

$= 5 + 11$

$= 16$

Q.16 Let $x^k + y^k = a^k$, $(a, k > 0)$ and $\frac{dy}{dx} + \left(\frac{y}{x}\right)^{\frac{1}{3}} = 0$, then k is

(1) $\frac{4}{3}$ (2) $\frac{2}{3}$ (3) $\frac{3}{2}$ (4) $\frac{1}{3}$

Ans. **[2]**

Sol. $x^k + y^k = a^k$

diff^n

$k.\, x^{k-1} + k\, y^{k-1} \frac{dy}{dx} = 0$

$\therefore \frac{dy}{dx} + \left(\frac{x}{y}\right)^{k-1} = 0$

on comparing $\frac{dy}{dx} + \left(\frac{y}{x}\right)^{1/3} = 0$

we get $k - 1 = -\frac{1}{3}$

$k = \frac{2}{3}$

Q.17 An unbiased coin is tossed 5 times. Suppose that a variable X is assigned the value k when k consecutive heads are obtained for k = 3, 4, 5, otherwise X takes the value –1. Then the expected value of X, is :

(1) $\frac{3}{16}$ (2) $-\frac{1}{8}$ (3) $\frac{1}{8}$ (4) $-\frac{3}{16}$

Ans. **[3]**

Sol.

k	0	1	2	3	4	5
p(k)	$\frac{1}{32}$	$\frac{12}{32}$	$\frac{11}{32}$	$\frac{5}{32}$	$\frac{2}{32}$	$\frac{1}{32}$

k : No of consecutive heads

Now, Expectation

$\Rightarrow (-1) \times \frac{1}{32} + (-1) \times \frac{12}{32} + (-1) \times \frac{11}{32} + 3 \times \frac{5}{32} + 4 \times \frac{2}{32} + 5 \times \frac{1}{32}$

$\Rightarrow \frac{4}{32}$

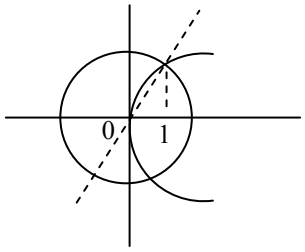
$\Rightarrow \frac{1}{8}$

Q.18 The area of the region, enclosed by the circle $x^2 + y^2 = 2$ which is not common to the region bounded by the parabola $y^2 = x$ and the straight line $y = x$, is :

(1) $\frac{1}{6}(24\pi - 1)$ (2) $\frac{1}{6}(12\pi - 1)$ (3) $\frac{1}{3}(12\pi - 1)$ (4) $\frac{1}{3}(6\pi - 1)$

Ans. **[2]**

Sol.



Required area = Area of circle $- \int_0^1 (\sqrt{x} - x)\ dx$

$= 2\pi - \int_0^1 (\sqrt{x} - x)\, dx$

$= 2\pi - \left(\frac{2x^{3/2}}{3} - \frac{x^2}{2} \right)_0^1$

$= \frac{12\pi - 1}{6}$

Q.19 If the system of linear equations
$2x + 2ay + az = 0$,
$2x + 3by + bz = 0$,
$2x + 4cy + cz = 0$
Where a, b, c $\in$ R are non-zero and distinct; has a non-zero solution, then :
(1) a, b, c are in G.P. (2) a, b, c are in A.P.
(3) $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are in A.P. (4) a + b + c = 0

Ans. **[3]**

Sol. For non-zero solution

$$\Delta = \begin{vmatrix} 2 & 2a & a \\ 2 & 3b & b \\ 2 & 4c & c \end{vmatrix} = 0$$

$2(3bc - 4bc) - 2a(2c - 2b) + a(8c - 6b) = 0$
$\Rightarrow ab + bc = 2ac$
$\Rightarrow$ a, b, c $\rightarrow$ H.P.
$\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ = A.P.

Q.20 A vector $\vec{a} = \alpha\hat{i} + 2\hat{j} + \beta\hat{k}$ $(\alpha, \beta \in R)$ lies in the plane of the vectors, $\vec{b} = \hat{i} + \hat{j}$ and $\vec{c} = \hat{i} - \hat{j} + 4\hat{k}$. If $\vec{a}$ bisects the angle $\vec{b}$ and $\vec{c}$, then :
(1) $\vec{a} \cdot \hat{i} + 3 = 0$ (2) $\vec{a} \cdot \hat{i} + 1 = 0$ (3) $\vec{a} \cdot \hat{k} + 2 = 0$ (4) $\vec{a} \cdot \hat{k} + 4 = 0$

Ans. **[Bonus]**

Sol. Angle bisector $\vec{a} = \lambda(\hat{b} + \hat{c})$ or $\mu(\hat{b} - \hat{c})$

$\vec{a} = \lambda \left(\frac{\hat{i} + \hat{j}}{\sqrt{2}} + \frac{\hat{i} - \hat{j} + 4\hat{k}}{3\sqrt{2}} \right)$

$= \frac{\lambda}{3\sqrt{2}} [4\hat{i} + 2\hat{j} + 4\hat{k}]$

comparing with $\vec{a} = \alpha\hat{i} + 2\hat{j} + \beta\hat{k}$

$\frac{2\lambda}{3\sqrt{2}} = 2$

$\Rightarrow \lambda = 3\sqrt{2}$

$\therefore \vec{a} = 4\hat{i} + 2\hat{j} + 4\hat{k}$ (Not satisfies option)

Q.21 $\lim\limits_{x\to 2} \dfrac{3^x + 3^{3-x} - 12}{3^{-x/2} - 3^{1-x}}$ is equal to

Ans. **[36]**

Sol. Given $\lim\limits_{x\to 2} \dfrac{3^x + 3^{3-x} - 12}{3^{-x/2} - 3^{1-x}}$

Let $3^{x/2} = t$

$\therefore \lim\limits_{t\to 3} \dfrac{t^2 + \dfrac{27}{t^2} - 12}{\dfrac{1}{t} - \dfrac{3}{t^2}}$

$\lim\limits_{t\to 3} \dfrac{(t^2-9)(t^2-3)}{(t-3)}$

$\lim\limits_{t\to 3} (t+3)(t^2-3)$

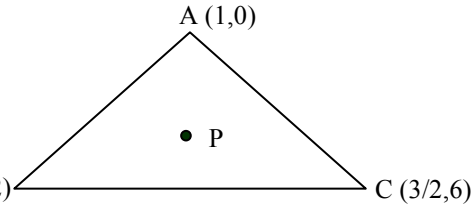
6×6

$\Rightarrow 36$

Q.22 Let A(1, 0), B(6, 2) and C $\left(\dfrac{3}{2}, 6\right)$ be the vertices of a triangle ABC. If P is a point inside the triangle ABC such that the triangles APC , APB and BPC have equal areas, then the length of the line segment PQ, where Q is the point $\left(-\dfrac{7}{6}, -\dfrac{1}{3}\right)$, is

Ans. **[5]**

Sol. Areas of triangles APC, BPC, APB are equal



therefore P is centroid

$p = \left(\dfrac{1+6+\frac{3}{2}}{3}, \dfrac{0+2+6}{3}\right)$

$= \left(\dfrac{17}{6}, \dfrac{8}{3}\right)$

$Q = \left(\dfrac{-7}{6}, \dfrac{-1}{3}\right)$

$PQ = \sqrt{\left(\dfrac{24}{6}\right)^2 + \left(\dfrac{9}{3}\right)^2}$

$= \sqrt{4^2 + 3^2}$

$= 5$

Q.23 If the sum of the coefficients of all even powers of x in the product $(1 + x + x^2 + \ldots.+ x^{2n})(1 - x + x^2 - x^3 + \ldots..+ x^{2n})$ is 61, then n is equal to

Ans. **[30]**

Sol. $(1 + x + x^2 + \ldots. x^{2n})(1 - x + x^2 - x^3 + \ldots x^{2n}) = a_0 + a_1 x + a_2 x^2 + \ldots..$

put x = 1

$(2n + 1) = a_0 + a_1 + a_2 + \ldots\ldots$ …(1)

put x = –1

$(2n + 1) = a_0 - a_1 + a_2 \ldots\ldots..$ ... (2)

add (1) & (2)

$4n + 2 = 2(a_0 + a_2 + \ldots..)$

$4n + 2 \Rightarrow 2 \times 61$

n = 30

Q.24 If the variance of the first n natural numbers is 10 and the variance of the first m even natural numbers is 16, then m + n is equal to …..

Ans. **[18]**

Sol. variance of (1, 2, ….. n)

$$\Rightarrow 10 = \frac{1^2 + 2^2 + \ldots.. + n^2}{n} - \left(\frac{1 + 2 + 3 + \ldots. + n}{n}\right)^2$$

on solving we get n = 11

Now,

variance of 2, 4, 6…….2m = 16

$\Rightarrow$ var (1, 2 ….. m) = 4

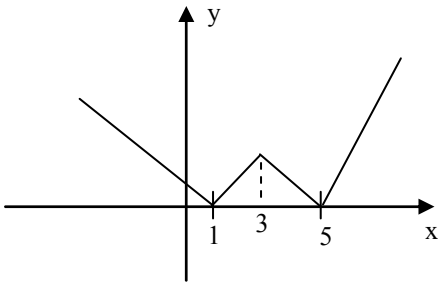
$\Rightarrow m^2 - 1 = 48 \Rightarrow m = 7$

$\therefore$ m + n = 18

Q.25 Let S be the set of points where the function, $f(x) = |2 - | x - 3||$, $x \in R$, is not differentiable. Then $\sum_{x \in S} f(f(x))$ is equal to ….

Ans. **[3]**

Sol. Graph of $y = |2 - |x - 3||$ is



Total points of N. D. are three

$\therefore$ S = 3 , 1, 5

$\therefore$ f(f(1)) + f(f(3)) + f(f (5))

$\Rightarrow$ 1 + 1 + 1

$\Rightarrow$ 3